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Cerebral hemodynamics and stroke risks in symptomatic intracranial atherosclerotic stenosis with internal versus cortical borderzone infarcts: A computational fluid dynamics study

Shuang Li, Xuan Tian, Bonaventure Ip, Xueyan Feng, Hing Lung Ip, Jill Abrigo, Linfang Lan, [Haipeng Liu](#), Lina Zheng, Yuying Liu, Yu Liu, Karen KY Ma, Florence SY Fan, Sze Ho Ma, Hui Fang, Yuming Xu, Alexander Y Lau, Howan Leung, Yannie OY Soo, Vincent CT MokKa Sing Wong, Xinyi Leng, Thomas W Leung

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Abstract

There may be different mechanisms underlying internal (IBZ) and cortical (CBZ) borderzone infarcts in intracranial atherosclerotic stenosis. In 84 patients with symptomatic, 50-99% atherosclerotic stenosis of M1 middle cerebral artery (MCA-M1) with acute borderzone infarcts in diffusion-weighted imaging, we classified the infarct patterns as isolated IBZ (n = 37), isolated CBZ (n = 31), and IBZ+CBZ (n = 16) infarcts. CT angiography-based computational fluid dynamics models were constructed to quantify translesional, post-stenotic to pre-stenotic pressure ratio (PR) in the MCA-M1 lesion. Those with IBZ infarcts were more likely to have a low PR (indicating impaired antegrade flow across the lesion) than those without (p = 0.012), and those with CBZ infarcts were more likely to have coexisting small cortical infarcts (indicating possible embolism) than those without (p = 0.004). In those with isolated IBZ or CBZ infarcts, low PR was independently associated with isolated IBZ infarcts (adjusted odds ratio = 4.223; p = 0.026). These two groups may also have different trajectories in the stroke risks under current medical treatment regimen, with a higher risk of same-territory ischemic stroke recurrence within 3 months in patients with isolated IBZ infarcts than isolated CBZ infarcts (17.9% versus 0.0%; log-rank p = 0.023), but similar risks later in 1 year.

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Keywords

- Borderzone infarct
- cerebral hemodynamics
- intracranial atherosclerotic disease
- ischemic stroke
- prognosis

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KW - cerebral hemodynamics

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